

## Notes from Network+ studies (foundational networking)

**Note:** The Network+ exam objectives are used to organize each section. Only information relevant to this lab iteration is included; unrelated objectives and concepts are omitted.

### 1.2 Networking Appliances

**Router:** A router is a Layer 3 device primarily used to connect different networks and forward traffic between them using IP addresses. Its main job is to determine the best path for packets to travel.

**Switch:** A switch is a Layer 2 device used to connect devices within the same local network/VLAN, using MAC addresses to forward frames to the correct device.

**Hub:** A hub is a Layer 1 device used to repeat incoming signals out to connected devices. It operates at Layer 1 because it only repeats electrical signals/bits out its other ports; it does not inspect or learn MAC addresses, and it does not make forwarding decisions.

**Firewall:** A firewall is a security device or software that controls network traffic based on configured rules, usually allow/deny rules. Firewalls commonly inspect source/destination IP addresses, and ports/protocols to decide whether traffic should be allowed or blocked. They can protect an entire network at the perimeter, sit between internal network segments, or run directly on an individual host/device (e.g., Microsoft Defender Firewall). A firewall can be a standalone appliance, built into a router, or installed as software on an endpoint. Basic firewalls mainly operate at Layers 3 and 4 (IP addressing security, and port/protocol security), while advanced firewalls can inspect Layer 7 application traffic.

### 1.2 Networking Functions

**Quality of Service (QoS):** QoS refers to the prioritization and management of certain types of network traffic over others. It is commonly used to give delay-sensitive traffic, such as VoIP or video conferencing, higher priority than less time-sensitive traffic, such as email, file downloads, or web browsing.

**Time to live (TTL):** TTL refers to a value given to packets which limits how long they can exist on a network. At each hop, the router decreases the value by 1; when it hits 0, the packet is dropped (prevents packets from looping forever).

### 1.4 Internet Protocol Types

**Internet Control Message Protocol (ICMP):** ICMP is used by network devices to send error messages and diagnostic information. It is commonly used by tools like ping and traceroute to test whether a device is reachable and how traffic moves through a network. Example: using ping to check if a server responds.

**Transmission Control Protocol (TCP):** TCP is a connection-oriented transport protocol used when reliable delivery matters. Before data is sent, TCP establishes a session between the two devices, then it tracks packets, acknowledges receipt, retransmits missing data, and delivers data in order. Its purpose is to make sure important data actually arrives correctly.

**User Datagram Protocol (UDP):** UDP is a connectionless transport protocol used when speed and low overhead matter more than guaranteed delivery. It sends data without first establishing a session and without waiting for acknowledgments. Its purpose is fast delivery with minimal delay.

### 1.4 Traffic Types

**Unicast:** Unicast is one-to-one traffic. One sender sends data to one specific receiver. This is the most common type of network traffic. An example would be your PC opening a website or connecting to a file server. Most normal client-to-server communication is unicast.

**Broadcast:** Broadcast is one-to-all within the local broadcast domain. One sender sends traffic to every device on that local network segment. An example would be ARP requests or older DHCP discovery traffic.

## 1.5 802.3 Standards and Types

When considering cabling in networking, it's important to be able to distinguish between the terminology. I like to separate physical networking into 3 layers of thought: physical media, protocol/technology type, and connectors/transceivers.

- Physical medium is what the signal physically travels through: Fiber carries light signals. Twisted-pair copper carries electrical signals. Coaxial cable carries electrical/RF signals. Twinax/DAC carries electrical signals over short copper connections.
- Protocol/technology type is what communication standard is being carried: Ethernet can run over twisted-pair copper, fiber, or DAC/twinax. Fibre Channel usually runs over fiber, but can also use DAC/twinax for short SAN connections. Coax is usually used for cable internet/cable TV, not normal switched Ethernet today.
- Connectors/transceivers are how the cable physically connects to equipment: Connectors really just determine how the cable physically connects (also useful for distinguishing cable types). Transceivers are essentially transmit/receive modules for a network port (I like to think of transceivers as translators, although that's probably not a perfect one to one analogy).

Example: A switch may not have built-in fiber ports, but it may have an SFP/SFP+ slot. You insert an optical transceiver into that slot, plug fiber into the transceiver, and the transceiver converts the switch's electrical signal into light for the fiber, then converts incoming light back into electrical signals the switch can process.

**Single-mode fiber:** Single-mode fiber uses a very small core and sends light in basically one path. Its purpose is long-distance, high-speed communication, such as between buildings, campuses, or ISP infrastructure. It commonly is used with LC connectors (also works with SC and ST).

**Multimode fiber:** Multimode fiber has a larger core and allows light to travel in multiple paths, which makes it better for shorter distances, like inside a building or data center. It commonly is used with LC connectors (also works with SC and ST). The practical difference is that single-mode is for longer range, while multimode is usually cheaper for short runs.

## 1.5 Wired networking protocols/standards

**Ethernet:** Ethernet is the main LAN networking standard used to carry normal network traffic between devices like PCs, switches, routers, servers, and access points. When you plug in a normal RJ45 Ethernet cable, that is Ethernet. When you plug in fiber or DAC between switches using SFP/SFP+/QSFP modules, that can also be Ethernet. Ethernet can run over copper twisted pair, fiber, or DAC/twinax depending on the port/transceiver/cable. Ethernet is general purpose networking. It is designed to carry many types of traffic: web traffic, file sharing, email, VoIP, streaming, cloud traffic, management traffic, etc. It is flexible, widely used, cheaper, and works great for normal LAN/WAN communication.

## 1.5 Transceiver/Connector Types

**Small form-factor pluggable (SFP)/(SFP+):** SFP is a small removable transceiver module used in networking appliances. It lets a device support different cable/media types, such as fiber or copper, without permanently building that connection type into the hardware. SFP supports speeds up to 1 Gbps, while SFP+ supports up to 10 Gbps – the tricky thing is SFP/SFP+ ports are near identical, so a SFP transceiver will usually work with a SFP+ port (bottlenecking it at 1 Gbps), however a SFP+ will not work in a designated SFP port because the SFP port lacks the hardware to process 10 Gbps.

**Lucent/Local Connector (LC):** LC is a smaller fiber optic connector that is very common in modern networking. It uses a latch mechanism and is often seen with SFP transceivers (Memory tip: LC = LatCh).

**Straight Tip (ST):** ST is a fiber optic connector that uses a twist-lock design, kind of like plugging in and turning it to lock. It is older and less common in modern Ethernet switching environments, but you may still see it in legacy fiber setups (Memory tip: ST = twist).

**Multi-fiber push on (MPO):** MPO is a fiber connector that holds multiple fiber strands inside one connector. Instead of connecting one fiber pair like LC or SC, MPO connects many fibers at once. It is used in high-density, high-speed environments such as data centers, especially for backbone links or 40G/100G+ connections. Commonly associated with QSFP style transceivers.

**RJ11:** RJ11 is a connector mainly used for telephone lines. It is narrower than RJ45 and usually has fewer contacts. It is commonly associated with voice lines, DSL, and older telecom connections rather than Ethernet LAN connections.

**RJ45:** RJ45 is the standard connector used for Ethernet over twisted-pair cabling, such as Cat5e, Cat6, and similar cables. It is the common connector on normal network cables plugged into PCs, switches, routers, and wall jacks.

## 1.7 Public vs. Private IPv4

**Public IP:** A public IP address is an address that is reachable on the internet. It is used so a device or, more commonly, a router/firewall can communicate with systems outside the local network. Example: if your home router connects to your ISP, the ISP gives it a public IP so your network can access the internet.

**Private IP:** A private IP address is an address used only inside a local network, such as a home, office, or school network. These addresses are not routable across the public internet, which means internet routers do not forward them publicly. Example: your laptop might have 192.168.1.25 inside your home network, while your router has the public IP that the outside world sees.

**Automatic Private IP Addressing (APIPA):** APIPA is when a device automatically gives itself an IP address because it could not get one from DHCP. In IPv4, this uses the 169.254.x.x range (169.254.0.0/16). Its purpose is to let a device still have a local address instead of having no usable IP at all. For example, if a PC cannot reach the DHCP server, Windows may assign it an APIPA address like 169.254.10.20. This solves the problem of a device being completely unaddressed, but it is limited because it usually only allows local network communication and not normal internet access.

**Loopback/localhost:** Loopback is a special IP address a device uses to talk to itself. In IPv4, localhost usually means 127.0.0.1. Its purpose is testing the local TCP/IP stack and local services without sending traffic onto the network. For example, pinging 127.0.0.1 checks whether your own machine's network stack is working.

## 2.2 Switching technologies and features

**Speed:** In this context, speed is just referring to the rate at which an interface sends and receives data.

**Duplex:** Duplex describes whether a network connection can send and receive data at the same time. **Half duplex** means an interface/port can send and receive, but not at the same time or a collision can happen (corrupting the frame). **Full duplex** means an interface/port can send and receive at the same time, which is the modern networking standard and avoids collisions on that link. Since duplex is configured per interface/port, both sides should agree on which duplex mode is used to avoid duplex mismatch issues.

**Maximum transmission unit (MTU):** MTU is the largest packet size that can be sent across a network link without being broken up into smaller pieces. Standard Ethernet commonly uses an MTU of 1500 bytes. If a packet is larger than the MTU, it may need to be fragmented or dropped, depending on the protocol/settings.

**Jumbo frames:** Jumbo frames are Ethernet frames with an MTU larger than the standard 1500 bytes, commonly around 9000 bytes. They are used to move larger chunks of data per frame, which can reduce overhead and improve efficiency (side note: jumbo frames only work properly if every device along the path supports and is configured for the larger MTU, if not, frames may be dropped or fragmented depending on the situation).

**Runts:** Runts are Ethernet frames that are smaller than the minimum valid Ethernet frame size, which is 64 bytes. They are usually considered malformed/corrupted frames and can indicate collisions, duplex mismatch, bad cabling, or faulty NIC/switch ports.

## 5.5 Networking device commands

**Show mac address-table:** Show mac address-table is entered on a switch, usually a managed Cisco switch or similar network switch CLI. It shows the switch's MAC address table, which maps learned MAC addresses to switch ports and VLANs. In simple terms, it tells you which device/MAC address the switch has learned on which port.

- Example use: if you want to know where a device is plugged in, you can look up its MAC address and see which switch port learned it.

**Show interface:** show interface is entered on a router, switch, firewall, or similar network device CLI. It shows detailed status and statistics for an interface, such as whether the interface is up/down, speed, duplex, MTU, bandwidth, input/output traffic, errors, drops, CRCs, runts, giants, and other interface counters. In simple terms, it tells you the health and activity of a specific port/interface.

- Example use: if a link is slow, unstable, or down, you can use show interface to check whether the interface is up, whether speed/duplex look correct, and whether error counters like CRCs, drops, runts, or giants are increasing.

**Show arp:** show arp is entered on a router, Layer 3 switch, firewall, or similar Layer 3 network device CLI. It shows the device's ARP table, which maps IPv4 addresses to MAC addresses for devices it has learned on connected networks. In simple terms, it tells you what MAC address the device knows for a given local IP address.

- Example use: if a device cannot reach another host or its gateway, you can use show arp to verify whether the network device has learned the correct IP-to-MAC mapping, or to troubleshoot duplicate IPs, missing ARP entries, or Layer 2 reachability issues.